

# Frame, Chebyshev, Conjugate Gradient algorithm

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## **Abstract**

This report examines signal reconstruction motivated by the frame reconstruction formula. Direct inversion of the frame operator  $S$  is impractical, which gives rise to the standard frame algorithm. To mitigate its slow convergence under large condition numbers, Chebyshev acceleration is adopted to speed up iterations. The conjugate gradient algorithm further addresses the difficulty of computing frame bounds  $A$  and  $B$ , allowing iterative reconstruction without explicit knowledge of these bounds. Nevertheless, several theoretical open problems remain for conjugate gradient and its practical applications. Further research is required to advance this line of work.

## Introduction

Many signal acquisition procedures produce a collection of inner-product coefficients, which act as measured information of an underlying signal. A central practical task is to reconstruct the original signal  $f$  solely from these coefficients. Frame theory was first formalized by Duffin and Schaeffer (1952) during their investigation of non-harmonic Fourier series and non-uniform sampling problems. They introduced the abstract definition of a frame for a Hilbert space: a sequence of non-zero vectors  $\{\varphi_i\}_{i=1}^n$  such that for any arbitrary vector  $f$  in the Hilbert space, there exist positive constants  $A, B > 0$ , that satisfies  $A\|v\|^2 \leq \sum_{i=1}^n \langle f, \varphi_i \rangle \langle \varphi_i, v \rangle \leq B\|v\|^2$ .

Frames enable stable non-orthogonal signal expansions. From the modern signal-processing viewpoint, the quantities  $\langle v, \varphi_i \rangle$  can be interpreted as physical measurements of  $f$ . Associated with every frame is the self-adjoint, positive-definite frame operator  $S$ , defined as  $Sv = \sum_{i=1}^n \langle v, \varphi_i \rangle \varphi_i$ .

Duffin and Schaeffer (1952) also constructed the conjugate frame  $\varphi_i' = S^{-1}\varphi_i$ , now widely referred to as the **canonical dual frame** in modern frame theory. The exact frame reconstruction formula reads  $v = \sum_{i=1}^n \langle v, \varphi_i' \rangle \varphi_i$ , which yields perfect signal recovery in theory. This identity is of great practical interest for signal processing, most notably for solving the irregular sampling problem.

While exact reconstruction is theoretically achievable via direct inversion of  $S$ , this approach faces critical numerical obstacles in real-world large-scale signal problems. Direct computation of  $S^{-1}$  becomes computationally expensive for high-dimensional signals. Furthermore, reconstruction quality degrades severely when the frame operator  $S$  possesses a large condition number (Christensen, 2003). These practical difficulties motivate the development of iterative reconstruction

algorithms. Three representative iterative approaches are considered in this report: the standard frame algorithm, Chebyshev acceleration, and the conjugate-gradient algorithm.

## Iterative Frame Reconstruction Algorithms

### Frame Algorithm

Duffin and Schaeffer (1952) presented the first prototype of a frame algorithm to approximate the original signal  $f$ . In this setting,  $A$  and  $B$  are bounds for  $\{\varphi_i\}_{i=1}^n$ . The constant  $\rho$  is a fixed relaxation parameter determined by these frame bounds:  $\rho = \frac{2}{A+B}$ . Setting this parameter, the iteration reads  $f_k = f_{k-1} - \rho \sum_{i=0}^n \langle f_{k-1}, \varphi_i \rangle \varphi_i$ ,  $f_0 = 0$ , with error estimate  $\|f - f_{k-1}\| \leq \left(\frac{B-A}{B+A}\right)^k \|f\|$ . The convergence rate in this algorithm is  $\frac{B-A}{B+A}$ . The further detailed steps can be seen in the work by Duffin and Schaeffer.

For sufficiently large iteration number  $n$ ,  $\left(\frac{B-A}{B+A}\right)^k$  will close to 0, which means that  $f_k$  will be close to original signal  $f$ . This provides a practical approach to reconstruct the original signal directly from frame coefficients, without explicit computation of the  $S^{-1}$ . This iterative scheme is also known as the Richardson iteration.

Nevertheless, this frame-based Richardson iteration suffers a critical practical limitation: its convergence performance relies on accurate estimates of frame bounds  $A$  and  $B$ . Exact values  $A$  and  $B$  are hard to get. Frequently, only a rough upper bound  $B$  or merely the existence of a lower bound  $A$  is available. Gröchenig (1993) pointed out that when only  $B$  is known, one may choose a corresponding relaxation parameter  $\rho = \frac{1}{B}$ . If neither  $A$  nor  $B$  is accessible, a sufficiently small relaxation parameter can be adopted as an approximate workaround.

Based on convergence rate  $\frac{B-A}{B+A}$ , it can be observed that convergence becomes extremely slow when

$B \gg A$ , corresponding to a large condition number of the frame operator  $S$ . In such cases,  $S$  is ill-conditioned.

Building upon the classical frame algorithm from Duffin and Schaeffer (1952), Jamali and Pourkani (2025) developed modified iterative reconstruction schemes for Hilbert-space frame approximation.

They derived a new recursive update formula

$$f_k = f_{k-1} + \frac{4}{A+B} \left( I_H - \frac{1}{A+B} S \right) S(f - f_{k-1}), \quad f_0 = 0, \quad \text{together with the error estimate is } \|f - f_{k-1}\| \leq \left( \frac{B-A}{B+A} \right)^{2k} \|f\|.$$

It shows that the new iteration possesses a squared convergence rate relative to the original frame algorithm. The method improves theoretical convergence speed for

poorly-conditioned frames, yet it still requires precomputed frame bounds  $A$  and  $B$ . This requirement motivates the polynomial acceleration methods discussed next.

### Chebyshev Algorithm

Gröchenig (1993) combined frame theory with polynomial-based acceleration techniques to develop the Chebyshev acceleration for frames. Unlike the original frame algorithm, this method explicitly incorporates known frame bounds  $A$  and  $B$  into the iteration scheme.

Given a frame  $\{\varphi_i\}_{i=1}^n$  with frame bound  $A, B > 0$ , and frame operator  $S$ , define  $\rho = \frac{B-A}{A+B}$ . The

Chebyshev recursion reads  $f_k = \lambda_n \left[ (f_{k-1} - f_{k-2}) + \frac{2}{A+B} S(f - f_{k-1}) \right] + f_{k-2}$ , where the relaxation parameter is updated recursively by  $\lambda_n = \frac{1}{(1 - \frac{\rho^2}{4})^n}$ , with  $\lambda_0 = 2, f_0 = 0$  and  $f_1 = \frac{2}{A+B} S f$ .

Introducing  $\sigma = \frac{\sqrt{B}-\sqrt{A}}{\sqrt{B}+\sqrt{A}}$ , we can obtain the error estimate:  $\|f - f_k\| \leq \frac{2\sigma^n}{1+\sigma^{2n}} \|f\|$ .

When the condition number of the frame operator is small, the convergence performance of

Chebyshev acceleration is similar to the basic frame algorithm. For sufficiently large condition

numbers, however, Chebyshev acceleration achieves substantially faster convergence. However, this approach requires a sufficiently accurate approximation of frame bounds  $A$  and  $B$ . If the estimations are not accurate, the convergence speed will be slow.

Adaptive Chebyshev schemes offer a practical alternative that can operate without exact prior knowledge of frame bounds  $A$  and  $B$ . Instead of requiring precise pre-computed spectral bounds, the algorithm starts from rough initial guesses and dynamically recalibrates its bound estimates by monitoring residual decay throughout iterations. Nevertheless, this adaptive mechanism relies on heuristic residual-based judgement rules rather than rigorous mathematical guarantees. When the actual residual decay deviates from theoretical expectations, the updated bounds may become unreliable, which harms convergence behaviour. Additional computational overhead is also introduced by repeated bound recalculation. Consequently, accurate frame-bound information remains a critical concern for Chebyshev-type methods, even in their adaptive variants (Hageman & Young, 1981).

Jamali and Pourkani (2025) proposed a modified Chebyshev-type iterative reconstruction scheme building on Gröchenig's (1993) acceleration work. They still use  $\sigma$  and  $\rho$ , but redefine them as  $\sigma = \frac{\sqrt{A^2 + B^2} - \sqrt{2AB}}{\sqrt{A^2 + B^2} + \sqrt{2AB}}$  and  $\rho = \left(\frac{B-A}{B+A}\right)^2$ . And the recursion formula is  $f_k = \lambda_n \left[ (f_{k-1} - f_{k-2}) + \frac{4}{A+B} S(f - f_{k-1}) \right] + f_{k-2}$ ,  $f_0 = 0$  with same error estimate formula.

Their analysis shows that this new  $\sigma$  is strictly smaller than the convergence constant obtained in Gröchenig (1993), yielding faster theoretical convergence. Nevertheless, this algorithm still requires exact prior knowledge of frame bounds  $A$  and  $B$ . This fundamental limitation of Chebyshev-based methods motivates the conjugate-gradient reconstruction approach.

## Conjugate Gradient Algorithm

To solve the problem that calculating the frame bound  $A, B$ , Gröchenig (1993) introduced a conjugate-gradient-based iterative algorithm for frame reconstruction. Starting from initial conditions  $f_0 = 0, r_0 = p_0 = Sf$ , the scheme recursively updates the signal approximation  $f_n$ , residual  $r_n$  and conjugate search directions  $p_n$ . The step-size parameter  $\lambda_n$  and new search directions are fully computed via inner-products within each iteration.

The approximate signal  $f_n$  is updated by  $f_n = f_n + \lambda_n p_n$ , which means that  $f_n$  can converge to original signal without knowledge of frame bound  $A, B$ .

The error estimate is  $\|f - f_k\| \leq \frac{2\sigma^n}{1+\sigma^{2n}} \|f\|$ , with  $\sigma = \frac{\sqrt{b}-\sqrt{a}}{\sqrt{b}+\sqrt{a}}$  and  $a, b$  are spectrum bounds of  $S$ .

Although missing actual frame bounds will affect the estimate of convergence speed, we can still converge to the original signal.

In contrast to the original frame algorithm and Chebyshev-type acceleration schemes, where frame bounds play a critical role in convergence performance, the conjugate-gradient-based reconstruction avoids the requirement for explicit frame-bound estimates. It mitigates convergence degradation caused by inaccurate bound estimation and offers a practical solution for scenarios where reliable frame bounds are hard to obtain. Furthermore, compared with those earlier methods, the conjugate-gradient approach requires significantly fewer iterations to reach the same target accuracy, especially when the frame operator has a large condition number

Given the widespread utility of the conjugate gradient method in solving symmetric positive-definite linear systems, further investigation into its practical convergence behavior remains necessary. van der Sluis and van der Vorst (1986) established the fundamental adaptive mechanism of CG, which

iteratively constructs a Krylov subspace and projects the original matrix to generate approximate smallest and largest eigenvalues known as Ritz values. In the finite frame setting, solving signal reconstruction problems essentially requires solving  $Sf_n = f$ .

Conjugate Gradient method utilizes Ritz values to approximate the true Maximum and minimum eigenvalues of the frame operator  $S$ , which are the frame bound A and B. However, this adaptive approximation mechanism leads to notable theoretical limitations. When Ritz values closely match the true eigenvalues of  $S$ , CG exhibits fast convergence and accurate error estimation. In contrast, slow early convergence arises if extremal Ritz values remain far from their target eigenvalues and fail to enter their final intervals. The error decay rate is also hard to calculate. The distribution of iteration polynomials  $R_i$  with respect to iteration coefficients is an open problem, and rigorous sharp quantitative error bounds are not available for arbitrary spectra. Many related observations can only be supported by numerical experiments.

## **Application**

Frame-based algorithms find broad applications across multiple settings, including Gabor frames, wavelet frames, irregular sampling, and trigonometric polynomials. Among these diverse topics, this report conducts more-detailed investigations focused on wavelet frames and irregular sampling.

## **Irregular sampling**

Irregular sampling acquires signal information at non-uniformly spaced sample points, and each sample can be interpreted as an inner-product-type measurement of the target signal. Duffin and Schaeffer (1956) established foundational results for perturbed oversampling, a form of irregular sampling. They demonstrated that under suitable conditions on sampling locations, a

norm-equivalence inequality holds:

$$A \int |f(x)|^2 dx \leq \sum |f(x_k)|^2 \leq B \int |f(x)|^2 dx, \text{ with sampling points } x_k.$$

Therefore, for an irregular-sampling problem, we aim to determine whether a band-limited function  $f$  can be fully reconstructed from its irregularly sampled values, and how this reconstruction can be achieved.

Since frame bounds  $A$  and  $B$  are typically unknown, it becomes difficult to recover the original signal  $f$  using frame-based approaches. The conjugate gradient method, which performs iterations without requiring explicit frame-bound knowledge, provides a powerful alternative. It enables signal reconstruction relying solely on available irregular-sampling measurements.

Feichtinger et al. (1993) employed conjugate-gradient iterations to accelerate iterative reconstruction for irregular sampling, avoiding the drawbacks of parameter-dependent Richardson iteration.

Integrating conjugate-gradient acceleration, adaptive weighting and Toeplitz matrix representations, they constructed the super-fast ACT algorithm suited for large-scale non-uniform sampling datasets.

Combining the conjugate gradient method with adaptive weighting for irregular sampling yields clear improvements in signal reconstruction. The conjugate gradient iteration accelerates the decay of reconstruction error compared with older fixed-relaxation iterations, while the adaptive weighting improves the conditioning of the underlying Toeplitz matrix. As a result, convergence is orders of magnitude faster and the number of floating-point operations drops substantially, compared with the earlier Marvasti method.

However, limitations remain. When the sampling set contains large gaps, even the weighted conjugate gradient scheme converges slowly, and only the preconditioned version can still recover the signal

under such difficult near-critical sampling conditions. In these cases, many more iterations are required. Moreover, these benefits rely heavily on the Toeplitz structure and the specific weighting scheme, so they are not guaranteed for arbitrary irregular-sampling problems. Plain conjugate gradient without adaptive weighting performs noticeably worse.

### Wavelet frame

A wavelet frame is constructed from a mother wavelet  $g \in L^2(R)$ , It is a powerful tool for analyzing non-stationary signals with time-varying frequency components, a limitation that standard Fourier analysis cannot handle well, since Fourier transform only provides global frequency information without local time resolution.

There are two types of wavelet-frame sampling patterns:

Regular sampling:  $\{\beta^{\frac{j}{2}}g(\beta^j x - \alpha k), j, k \in Z\}$ , with fixed lattice parameters  $\alpha > 0, \beta > 1$ .

Irregular sampling:  $\{\beta_n g(\beta_n(x - x_n)), n \in Z\}$ , where  $\{(x_n, \beta_n)\}$  are arbitrary sampling points taken from the upper half-plane.

For regular wavelet frames, the frame-bound ratio  $B/A$  can be close to 1. Under this condition, the standard frame reconstruction algorithm or Chebyshev acceleration can work effectively.

In contrast, for irregular sampling configurations, the ratio  $B/A$  becomes very large. This leads to a large number for frame operator  $S$ . Consequently, the conjugate gradient method becomes useful for stable reconstruction in this setting (Grochenig, 1993).

Law and Liew (1997) applied the conjugate gradient method to wavelet-based signal reconstruction.

They found that conjugate gradient method yields equivalent results to the accelerated frame algorithm

originating from frame theory. However, the paper does not further discuss the practical performance and potential pitfalls of deploying conjugate gradient algorithm within wavelet-frame settings.

### **Unsolved Problems and Research Gaps**

The distribution of iteration-dependent polynomial  $R_l$  within conjugate gradient iterations remains an open theoretical problem. Rigorous quantitative proofs for this quantity are still absent, which obstructs sharp error-bound derivations for CG reconstruction. Although CG can perform signal reconstruction and achieve convergence without explicit knowledge of frame bounds A and B, reliable error estimation and convergence-precision prediction still rely on these frame-bound quantities. Therefore, frame-bound information remains critical for faithful original-signal recovery.

CG also suffers from inherent limitations tied to the behaviour of Ritz values. When extremal Ritz values deviate substantially from the true eigenvalues of the frame operator S, early-stage convergence slows down, and formal error analysis becomes difficult. Wavelet frames are valuable for non-stationary signals with time-varying frequency components, and irregular sampling scenarios are frequently encountered in practice, which makes CG an indispensable numerical tool for wavelet-frame reconstruction. Nevertheless, literature treating the combination of wavelet frames and conjugate gradient methods remains limited. Law and Liew (1997) combined CG with wavelet-based reconstruction, yet their work did not address these theoretical drawbacks. Consequently, further investigation into the coupling of wavelet-frame theory and the conjugate gradient algorithm is still needed.

## Conclusion

This report traces the origin of the frame reconstruction formula, which motivates the exploration of signal-recovery approaches built upon this formulation. The standard frame algorithm suffers from slow convergence when the frame operator  $S$  has a large condition number; additionally, the frame bounds  $A$  and  $B$  are often not readily computable in practice. To address the slow-convergence issue under high condition numbers, Gröchenig (1993) developed the Chebyshev acceleration algorithm. Though this method still requires known frame bounds  $A$  and  $B$ , it yields substantially faster convergence for ill-conditioned operators. To resolve the problem of unknown  $A$  and  $B$ , Gröchenig (1993) further introduced the conjugate gradient algorithm, which iteratively approximates the original signal without explicit frame-bound knowledge. Nevertheless, without known frame bounds, accurate error estimation becomes difficult, leaving this as an open direction for future investigation. This report further extends the discussion to Chebyshev and CG algorithms applied to irregular sampling within wavelet-frame settings. While Law and Liew (1997) implemented CG for wavelet-maxima reconstruction, deeper theoretical study integrating CG and wavelet-frame theory remains another promising problem for further research.

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